

AI Investment and Market Competitiveness

The Critical Role of Human-Machine Interaction and Production Efficiency



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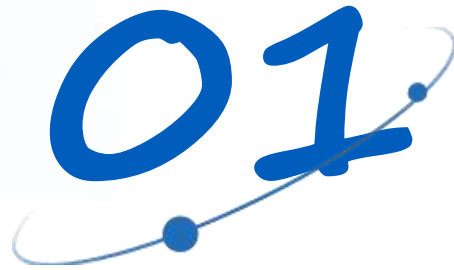
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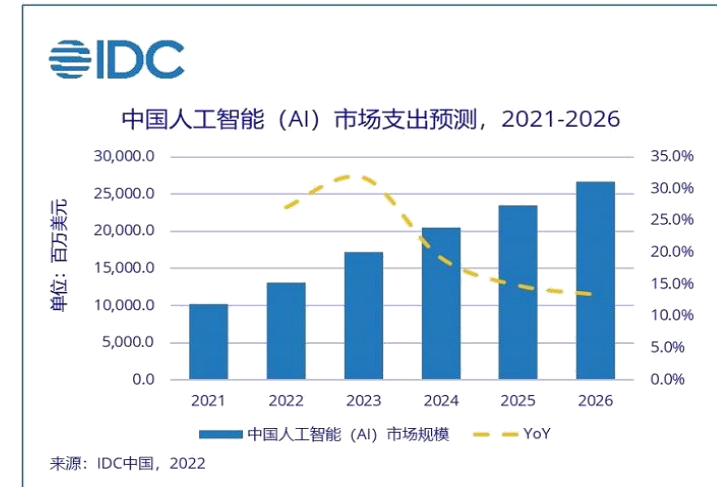


Introduction & Research Gap

Introduction



Global AI Investment Trend



Despite the huge investment in AI, there is a significant differentiation in the competitive outcomes among enterprises.

How (and when) can AI investment truly translate into market competitiveness?





Limitations of existing literature

- ✓ Academic studies have mainly linked AI to innovation/financial returns
- ✗ the internal transmission channel from AI investment to market competitiveness unexplored



Unclear Employee Roles

- ✗ Employees, as the core of execution, have been underestimated
- ✗ the boundary role of employees' **skills and training in the "technology-efficiency-competitiveness" path** has not been quantified.

GAPS



RESEARCH TARGETS

Construct a Comprehensive Framework with **Production Efficiency** as the Mediator and **Employee Training** as the Moderator

Empirically Test the 3 Core Questions:

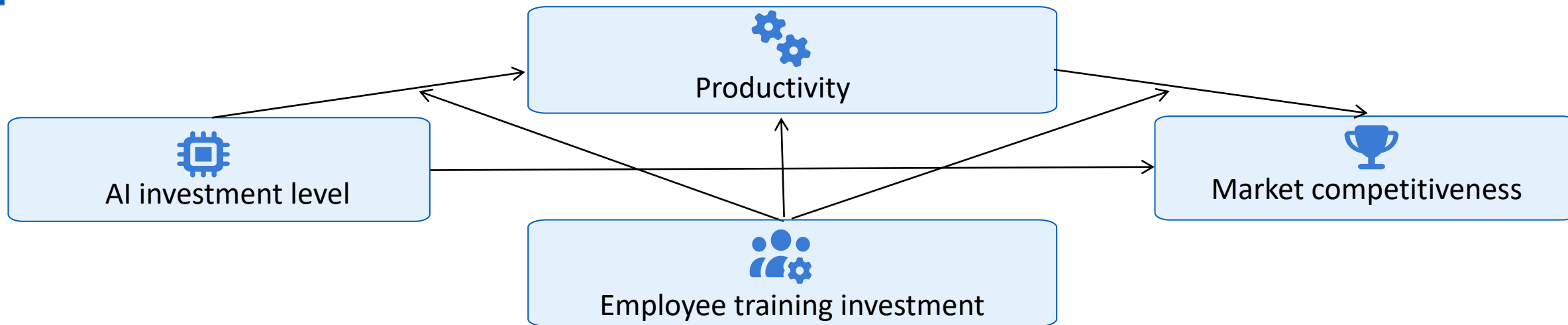
1. How Does AI Investment Directly Impact Market Competitiveness?
2. Does Production Efficiency Play a Mediating Role Between the Two?
3. Does Employee Training Moderate the Impact Path of AI Investment?



02

Hypotheses

Conceptual Framework



Hypotheses

- ❑ H1: AI investment is positively associated with market competitiveness.
- ❑ H2: Production efficiency mediates the relationship between AI investment and market competitiveness.
- ❑ H3: Employee training investment strengthens the positive effect of AI investment on production efficiency.
- ❑ H4: Employee training moderates the mediated relationship between AI investment and market competitiveness via production efficiency.

03

Methodology

Sample source and measurement



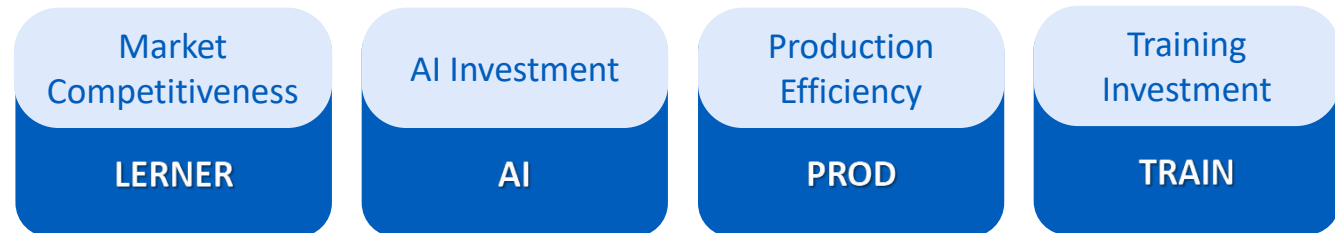
Sample source

Manufacturing companies
listed on the A-share market
from 2020 to 2023

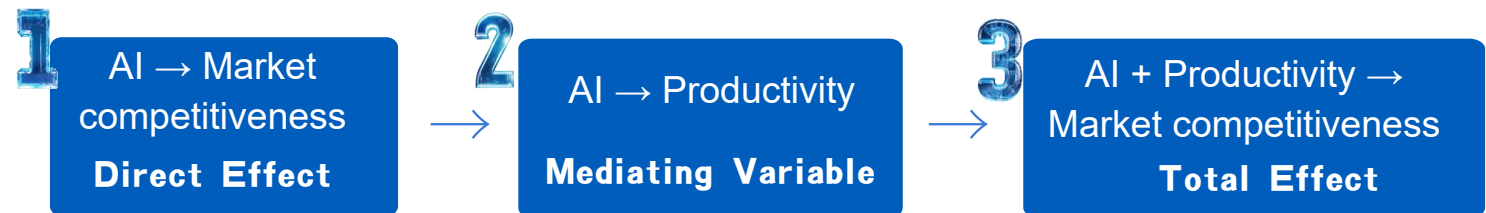
9,124
firm-year observations

Measurement of Core Variables

Control variables include size, leverage, age, ownership, with year and firm fixed effects controlled.



Models



A stepwise OLS regression is employed, with all continuous variables winsorized at the 1% level.
Year and firm fixed effects are controlled for, and standard errors are clustered at the firm level to ensure robust results.

Variable	Min.	Max.	Mean	Stdv.
LERNER	-0.349	0.532	0.133	0.134
PROD	0.704	0.916	0.771	0.037
AI	0.000	0.038	0.004	0.006
TRAIN	0.000	0.720	0.143	0.142
SCALE	19.437	26.721	22.356	1.414
DEBT	0.069	0.913	0.421	0.191
AGE	11.016	38.537	22.688	5.453
SOE	0.000	1.000	0.259	0.438

	Model (1) LELRNER	Model (2) PROD	Model (3) LELRNER
Con_	15.710*** (0.340)	133.94*** (0.826)	12.170*** (0.518)
AI	-2.100** (-0.613)	-1.700* (-0.142)	-2.360** (-0.689)
PROD	-	-	-5.090*** (-0.234)
TRAIN	-2.110** (-0.024)	-17.810*** (-0.059)	-7.140*** (-1.448)
AI*TRAIN	1.960** (2.240)	2.140** (0.698)	2.340** (2.663)
PROD*TRAIN	-	-	7.020*** (1.860)
SCALE	-3.180*** (-0.003)	-6.990*** (-0.002)	-3.090*** (-0.003)
DEBT	-36.110*** (-0.255)	-14.070*** (-0.028)	-35.480*** (-0.253)
AGE	-4.100*** (-0.001)	2.090** (0.000)	-4.220*** (-0.001)
SOE	-1.92* (-0.006)	0.210 (0.000)	-1.500 (-0.005)
Year/Individual	CONTROL	CONTROL	CONTROL
F	148.800***	64.700***	130.460***
R sq.	0.152	0.072	0.157



04

Key Findings

Direct Effect: Negative



AI investment alone shows a significantly negative impact on both market competitiveness and production efficiency

validating the short-term cost pressure hypothesis.

- AI boosts market position only if it optimizes production
- without this, high initial investments raise costs, not returns.

Conclusion

Discoveries

Mediating Effect: Partially Positive



After introducing production efficiency, the negative effect of AI on competitiveness through the efficiency pathway is attenuated

indicating that efficiency plays a partial mediating role.

Moderating Effect

**AI × Training
→ Productivity**

Evidence of Employee
Training's Moderating Effect:
Reversing Negative Impacts

**AI × Training
→ Competitiveness**

The interaction term similarly demonstrates significantly positive effects, with training directly amplifying AI's positive influence on market performance.

**Productivity × Training
→ Competitiveness**

Training further strengthens the positive correlation between efficiency and competitiveness, creating synergistic benefits.

Conclusion

Adequate investment in human capital can fully unleash AI's technological potential, forming a synergistic "technology + talent" enhancement effect.



Conclusion & Implications

Conclusion



Short-Term Costs

AI investment alone can erode competitiveness and efficiency in the short run due to integration friction and cost burdens.



Efficiency as a Conduit

Production efficiency serves as a crucial yet partial conduit for translating AI investment into market gains.



Training as a Boundary Condition

Employee-training spending constitutes a positive boundary condition that redirects AI's impact toward favourable territory.

CORE INSIGHT

The realization of AI value depends on human-machine collaboration.

Neglecting complementary human capital investment will lead to technology conversion failure.

Enterprise Strategy and Policy Suggestions

Managerial Actions

■ Phased Advancement

Avoid overinvestment and plan technology and talent simultaneously.

■ Precision Training

Focus on the collaborative skills of data analysis and intelligent manufacturing.

■ Strengthen Synergy

Build a human-machine collaboration mode and enhance organizational learning capacity.

Policy Actions

■ Policy Guidance

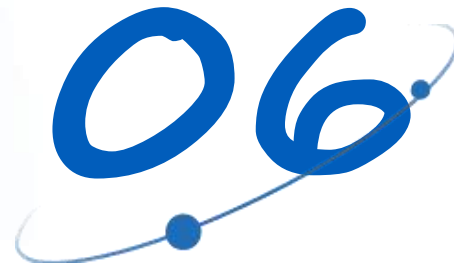
Improve fiscal subsidies and guide enterprises to invest rationally.

■ Public Platform

Support industry – university – research cooperation and establish AI skill re-education.

■ Public Platform

Establish data sharing and standards to lower the threshold for small and medium – sized enterprises.



Limitations & Future Research

Limitations & Future Research

Limitation



Sample Limitation: The study focuses solely on A-share listed manufacturing enterprises, requiring caution when generalizing the findings.



Variable Limitation: adopts a proxy variable that aligns with mainstream practices in existing literature

Future Directions



Expand Sample Scope: Include multi-country and multi-industry subjects to enhance the generalizability of conclusions.



Test Robustness: Explore more direct or refined approaches to measure AI investment.

THANK YOU FOR WATCHING



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